

Number of Partial-FFT Views

9 August 2026

1 Introduction

Underwater acoustic orthogonal frequency division multiplexing (OFDM) receivers operate on channels that are time varying and frequency selective. Partial-FFT reception keeps time-local components of the full-interval fast Fourier transform (FFT), while combiner weights form one pre-decoder soft symbol per carrier.

We reconsider whether four partial-FFT views help at $N = 1024$, why the pilot-only Partial-FFT receiver can lose to OFDM with forward error correction (FEC), and whether the result depends on the code rate. We separate three kinds of statements: mathematical identities, measurements in the retained files, and possible explanations that require another comparison.

All receivers at one printed configuration receive the same waveform. Hence their uncoded and coded transmission rates are identical before receiver errors are counted. Receiver performance changes the effective data rate through

$$R_{\text{eff}} = \frac{B_{\text{pay}}}{T_{\text{frame}}} \text{PSR}, \quad (1)$$

where B_{pay} is the payload bits per frame, T_{frame} is the frame transmission duration, and PSR is the payload-exact frame success rate. This expression separates waveform rate from the fraction of frames recovered by a receiver.

The two result families answer different questions. Table 1 records the boundary used throughout this note.

Table 1: Result families used in this note.

Result family	Channel and added noise	Frame use	Comparison supported
First-four-second AWGN	Measured Red replay with complex Gaussian noise that is independent over samples	Four approximately non-overlapping frames per path and SNR, one seed	Receiver counts at one fixed waveform with four views
Legacy added- noise sweep	Measured Red replay with impulsive noise correlated across the three hydrophones	Sixty frames per path and SNR, one seed; the same workloads recur across SNR after noise scaling	Three matched directory pairs with one and four views at code rate 0.125
Selected $N = 1024$, CP= 128 rate rows	Legacy sweep with impulsive added noise	Sixty frames per path and SNR, one seed	Code rates 0.25 and 0.5 represented only by one-view runs in this selection

The first family is additive white Gaussian noise (AWGN), whereas the matched view comparison is not. We therefore do not use the legacy comparison to identify the cause of the AWGN panel. The historical rate-0.5 rows are not part of the retained AWGN results.

2 OFDM Demodulation and Partial-FFT Combining

2.1 Partial-FFT Views

We first state what changing the number of views cannot change. Let $r[n]$ be the useful received interval and let $g_m[n]$ select view m . We write the carrier- k view as

$$Y_{m,k} = \sum_{n=0}^{N-1} g_m[n] r[n] e^{-j2\pi kn/N}, \quad (2)$$

where N is the useful interval length, n is the sample index, k is the carrier index, and m is the view index. Each view retains the contribution from one non-overlapping part of the useful interval.

The implementation windows form a partition of the useful samples:

$$\sum_{m=1}^M g_m[n] = 1, \quad (3)$$

where M is the number of partial-FFT views. Linearity of the FFT then gives

$$\sum_{m=1}^M Y_{m,k} = Y_k^{\text{full}}, \quad (4)$$

where Y_k^{full} is the full-interval FFT output on carrier k . Equation (4) holds for every number of non-overlapping views used by this receiver.

The OFDM+FEC receiver applies residual pilot equalization after the sum in (4). It must therefore be independent of the configured number of views, apart from numerical precision. Across each of the three legacy pairs, its successes, bit errors, bit error rate (BER), and effective rate match exactly in all 192 path-SNR cells.

This agreement is strong regression evidence, but the result file does not prove the identity. The identity follows from the partition and FFT linearity; a numerical test protects its implementation.

2.2 The Pilot-Trained Combiner

The pilot-only receiver does not use the sum with equal weights. For carrier k in band b , it forms

$$Z_k = \sum_{m=1}^M w_{m,b} Y_{m,k}, \quad (5)$$

where $w_{m,b}$ is the complex weight for view m and carrier band b . Unequal weights preserve a different combination of the time-local components than the full FFT.

The weights solve a bandwise ridge problem:

$$\hat{\mathbf{w}}_b = \arg \min_{\mathbf{w} \in \mathbb{C}^M} \left\{ \sum_{p \in \mathcal{P}_b} |\mathbf{w}^T \mathbf{Y}_p - X_p|^2 + \delta \|\mathbf{w}\|_2^2 \right\}, \quad (6)$$

where $\mathbf{Y}_p = [Y_{1,p}, \dots, Y_{M,p}]^T$, $\mathbf{w}$ is a trial direct coefficient vector, $\hat{\mathbf{w}}_b$ is the fitted vector, X_p is the known symbol, δ is the ridge term, and $\|\cdot\|_2$ is the Euclidean norm. The set $\mathcal{P}_b$ contains the pilot targets in band b ; when the band contains fewer than M targets, the implementation uses all target pilots. The transpose in (6) matches the direct coefficient sum in (5). This fit trades pilot error against the size of the combiner weights.

After combining, the receiver divides every pilot by its known symbol, interpolates the resulting response over the active carriers, and divides Z_k by that response. With one view and one global band, the fitted combiner is one scalar α , giving

$$Z_k = \alpha Y_k^{\text{full}}, \quad \hat{H}_k = \alpha \tilde{H}_k, \quad \frac{Z_k}{\hat{H}_k} = \frac{Y_k^{\text{full}}}{\tilde{H}_k}, \quad (7)$$

where $\tilde{H}_k$ is the response obtained by interpolating the unscaled pilot responses. The fitted scalar cancels, provided it does not reach the implementation's guard for a response near zero.

The deployed receiver uses 16 bands. A different scalar can therefore be fitted in each band, and pilot interpolation can cross a boundary between two scalars. With four views, the weighted result is generally not a scalar multiple of the full FFT, so the cancellation in (7) does not apply.

The current package test checks only that one raw view equals the full useful interval FFT. It does not check equality after combining one view with one band spanning all active carriers and applying residual pilot equalization. Thus the proposed PFFT-002 test is mathematically sound, but it is not yet present in `Pkg.test()`.

3 Measured-Channel Results

We first examine the panel that motivated the question. It uses $N = 1024$, CP=64, code rate 0.25, pilot spacings 5/5, check degree 14, and four views. At 30 dB, pooling the twelve Red channel-hydrophone paths gives Table 2.

Table 2: First-four-second AWGN frame counts at 30 dB.

Receiver	Successful frames	Frame success rate
OFDM+FEC	42/48	0.875
Pilot-only Partial-FFT+FEC	36/48	0.750
JUNA-Lite	46/48	0.958
JUNA (C, z) experiment receiver	46/48	0.958
JUNA (C, W, z) experiment receiver	46/48	0.958

The count of four frames per path is too small to order the receivers. The OFDM and JUNA-Lite totals differ by four frames, and the saved files do not retain their paired frame outcomes. We therefore report the counts without claiming an improvement for receivers using the same waveform.

The matched legacy comparison is larger but belongs to the impulsive added-noise family. Table 3 pools SNR from 20 to 30 dB, giving 4320 frame trials represented by aggregate counts per receiver and column. Four views reduce the pilot-only receiver's pooled success rate in all three pairs, while they increase the JUNA-Lite rate. The (C, z) and (C, W, z) experiment receivers differ from JUNA-Lite by no more than 0.0031 in these comparisons with four views. The gradient-only receivers therefore do not materially change the recorded pooled rates; the large increase is already present in JUNA-Lite.

The table does not constitute a four-view comparison with equal weights. Changing from one to four views changes the dimension of the ridge fit, the view signals, and the stopping behavior of JUNA-Lite. A receiver with four views and one common fixed weight is required to isolate the decomposition itself.

Table 3: Legacy frame success rates with impulsive added noise and four or one partial-FFT views.

Geometry	Receiver	Four views	One view
CP16, pilots 3/5	OFDM+FEC	0.827778	0.827778
	Pilot-only Partial-FFT+FEC	0.566435	0.816204
	JUNA-Lite	0.930787	0.815509
CP16, pilots 3/10	OFDM+FEC	0.811343	0.811343
	Pilot-only Partial-FFT+FEC	0.546296	0.802083
	JUNA-Lite	0.931481	0.803241
CP128, pilots 5/10	OFDM+FEC	0.698843	0.698843
	Pilot-only Partial-FFT+FEC	0.371991	0.674769
	JUNA-Lite	0.896991	0.677315

The available rows do not separate code rate from the number of views. Table 4 shows the settings behind the earlier interpretation that the result depends on code rate.

Table 4: Historical high-SNR comparison at $N = 1024$ and CP= 128.

Code rate	Views	OFDM+FEC	Best JUNA receiver	Reading
0.125	4	0.698843	0.897917	Four-view result
0.25	1	0.735880	0.725926	One-view result
0.5	1	0.176620	0.162269	One-view legacy result

The table supports a gain with four views in the tested rate-0.125 legacy configuration. It does not show that the gain is confined to that code rate, because the other code rates were measured with one view. Failure to measure a difference is also not evidence that two receivers are equivalent.

4 Limitations

Every receiver within one run receives a copy of the same received waveform, so the receiver outcomes are paired. The calculation that treats the two receiver proportions as independent gives $z = 1.477$ and $p = 0.140$ for the AWGN counts. However, that calculation does not match the experiment. From the two margins alone, the possible exact two-sided exact two-sided McNemar p -values range from 0.125 to 0.289.

The legacy calculations that give p -values below 10^{-50} also treat 4320 outcomes per receiver as independent. However, each SNR uses the same sixty frame seeds, the receivers share each received waveform, and the three hydrophones share correlated added noise. With four capture groups, the minimum attainable two-sided exact sign-test p -value is 0.125.

We therefore use the legacy differences as descriptions of these retained files. Repeated seeds support statements about these captures, while more independent captures are required for statements beyond the retained Red files. Both cases require frame traces for every receiver and a paired comparison that keeps the capture as a group. Overlapping marginal intervals do not replace that paired comparison.

5 What Each Receiver Contributes

The retained results identify several effects but do not select one cause. We therefore state each explanation with the measurement that would oppose it.

At $N = 1024$, pilot spacing 5 gives 205 pilots, or only 12–13 pilots in each of 16 bands. The pilot-only receiver estimates four complex weights per band from these targets. At $N = 2048$ and $N = 4096$, the corresponding counts are 25–26 and 51–52 pilots per band.

The four columns of the ridge problem can be correlated because every view comes from the same received symbol. A poorly conditioned correlation matrix can make the fitted weights sensitive to pilot noise, even when the ridge term keeps the solve finite. This explanation predicts large condition numbers and large weight norms where pilot-only Partial-FFT reception loses.

For uncorrelated view noise, the combined noise variance is

$$\sigma_Z^2 = \sum_{m=1}^M |w_{m,b}|^2 \sigma_m^2, \quad (8)$$

where σ_m^2 is the noise variance in view m and σ_Z^2 is the variance after combining. Large fitted weights can therefore increase the noise passed to the decoder.

The implementation did not retain the weights, correlation matrix condition numbers, or the noise gain. The explanation based on the number of pilots is therefore consistent with the N crossover, but the current files do not establish it.

One weight vector is shared by every carrier in a band. Residual pilot equalization then interpolates pilot responses across active carriers, including carriers near a band boundary. Different weight vectors on the two sides need not pass through that interpolation as one common factor.

This explanation predicts a smaller gap with one band spanning all carriers, with weights that vary continuously across frequency, or with the same fixed weights in all bands. It also explains why the identity for one view and one band spanning all carriers says nothing about the deployed 16-band path.

5.1 OFDM Under Residual Doppler

Shorter FFT intervals reduce mixing among carriers, and weighted combining of their outputs can reduce intercarrier interference (ICI) [1]. In the notation of this note, equal weights reconstruct the full FFT, while unequal weights retain a different combination of the view coefficients.

At a sampling rate of 9.6 kHz, the useful interval and one of four views have the durations in Table 5.

Table 5: Useful-interval and single-view durations for four views at 9.6 kHz.

N	Useful interval	One of four views
1024	106.7 ms	26.7 ms
2048	213.3 ms	53.3 ms
4096	426.7 ms	106.7 ms

On a time-varying channel, longer symbols can expose more within-symbol variation to the views. However, larger N also supplies more pilots per band, so the measured crossover cannot be assigned to symbol duration alone. A comparison with the channel held fixed and another with the same number of pilots are both required.

5.2 Candidate Selection and Stopping

The frame-wide JUNA-Lite path starts from the pilot-only Partial-FFT candidate. It returns immediately when that candidate satisfies the low-density parity-check (LDPC) syndrome. In this path, the validity flag means a zero syndrome and does not mean that a cyclic redundancy check (CRC) passed. Four-view pilot-only reception performs poorly in the three legacy pairs, so it can cause JUNA-Lite to enter posterior-anchor refinement more often. The pilot-only reception with one view performs much better and can stop refinement earlier. The measured JUNA-Lite difference may therefore combine useful time-local information with a change in how often refinement runs.

This explanation predicts that the gap changes when both view settings run the same fixed number of refinements. It also predicts parity-valid but CRC-invalid early returns if a wrong codeword satisfies the parity checks. The saved Lite files do not retain these events.

5.3 LDPC Posterior Symbols

A lower code rate supplies more redundant bits and can give the decoder more reliable posterior information for the JUNA-Lite anchor set. Decoder feedback is useful when pilots alone do not specify stable combiner weights and when the posterior information is reliable.

This explanation predicts a larger Lite gain with four views at lower code rate after the number of views, noise realization, geometry, and stopping rule are held fixed. The existing rate table does not make that comparison, so the explanation based on code rate remains untested.

5.4 Cyclic Prefix

The cyclic prefix (CP) reduces intersymbol interference when it covers the channel delay spread. For each path in the audit, squared tap magnitudes were averaged over replay snapshots to form a mean power-delay profile. The shortest contiguous interval containing 90 percent of its energy ranges from about 3.1 to 8.1 ms. CP64 is 6.67 ms and covers eleven of the twelve mean windows under that rule, while CP128 is 13.33 ms and covers all twelve.

The 90-percent rule is a design choice rather than a physical boundary. A shorter CP leaves residual energy outside the guard, while a longer CP spends more transmission time on the guard. CP128 is therefore a conservative guard under this rule, not the only usable guard. The audit calculation does not yet have a tracked script and is not a reproducible result until that script is retained.

A coherence calculation using the tap estimates alone also omits the separate samplewise phase trajectory used by the replay channel. The quoted coherence time therefore cannot be used as a hard limit that discards every longer- N result. Four-view reception is intended to retain information when the channel varies within the full useful interval.

6 Complexity and Evaluation

The simple OFDM+FEC receiver does not run the Partial-FFT combiner refinement. Both receivers can stop after one decoder run when the initial JUNA-Lite candidate is valid. JUNA-Lite adds combiner fits and decoder runs only when refinement proceeds. At one waveform, it cannot increase the transmitted raw rate; it can increase the delivered rate only by recovering more frames.

The present Red evidence does not support the earlier statement that JUNA should not be used. It supports a narrower statement: JUNA-Lite with four views has a large descriptive gain in three rate-0.125 legacy comparisons with impulsive added noise, while the AWGN comparison with four frames per path remains inconclusive.

One proposed policy would decode with Standard first, let a failed CRC trigger JUNA-Lite, use CRC for Lite stopping and final selection, and return an erasure if Lite also fails. That policy was not measured. The AWGN (C, z) receivers set `cz_crc_gate=false` and `cz_gradient_only=true`, while frame-wide JUNA-Lite starts from pilot-only Partial-FFT reception and stops on LDPC validity rather than CRC validity. The measured receivers do not demonstrate a CRC-gated policy.

7 Experimental Method

We can distinguish the explanations with the controlled comparisons in Table 6. Each comparison keeps the received frames, pilot symbols, code, channel, and added noise fixed unless the named quantity is the one being changed.

Table 6: Controlled comparisons for the remaining explanations.

Question	Controlled comparison	Evidence to retain
One-view cancellation	One view, one global band, and the same input through Standard and Partial-FFT equalization	Equalized arrays before decoding, with a numerical tolerance
Four-view reconstruction	Four views with every weight fixed to the same global value, against Standard	Equalized arrays, BER, and frame success
Band-boundary effect	One band against 16 bands with the same four views and targets	Per-band weights and carrier error near each boundary
Weight-estimation loss	Pilot-only reception with four views and logged ridge matrices	Condition numbers, weight norms, pilot residuals, and noise gain
Lite stopping effect	One and four views with the same fixed number of Lite refinements	Initial syndrome, CRC, selected iteration, and every candidate outcome
Code-rate effect	Matched AWGN runs with one and four views at rates 0.125 and 0.25	Paired frame outcomes for every receiver; rate 0.5 only if that experiment is reopened
Channel-variation effect	Frozen taps and phase against the measured replay, with pilot support held fixed	Effective-channel variation, weights, BER, and frame success
Deployment policy	Standard first, followed by CRC-gated Lite only after failure	Effective data rate, processing latency, decoder runs, and undetected errors

The frame traces must retain Standard, pilot-only Partial-FFT, and JUNA-Lite, not only the gradient-only receivers. Repeated seeds and additional independent captures are needed before probabilities are used to generalize beyond the retained Red files.

We record the branches, exact experiment directories, result manifests, and receiver files so that each statement can be checked.

- AWGN panel: branch `agent/awgn-results`, decision AWGN-018, and manifest `JunaCore/experiments/2026-08-08-red-awgn-first4s-frames4-snr-sweep-n1024-cp64-rate025-p5-5-dc14-kfill-pfft4/results/results_manifest.json`. Its configuration is $N = 1024$, CP= 64, rate 0.25, pilots 5/5, check degree 14, and four views. The manifest lists the twelve configuration CSV files used for Table 2.
- Legacy matched pairs: branch `agent/all-results`. Under `JunaCore/experiments`, the exact directory

pairs are

- 2026-08-06-red-snr-sweep-n1024-cp16-rate0125-p3-5-dc10-kfill and 2026-08-06-red-snr-sweep-n1024-cp16-rate0125-p3-5-dc10-kfill-pfft1;
- 2026-08-06-red-snr-sweep-n1024-cp16-rate0125-p3-10-dc10-kfill and 2026-08-06-red-snr-sweep-n1024-cp16-rate0125-p3-10-dc10-kfill-pfft1;
- 2026-08-06-red-snr-sweep-n1024-cp128-rate0125-p5-10-dc10-kfill and 2026-08-06-red-snr-sweep-n1024-cp128-rate0125-p5-10-dc10-kfill-pfft1.

Each directory contains `results/results_manifest.json`; each manifest lists the twelve configuration CSV files used for Table 3.

- AWGN receiver provenance: the result manifest pins these values:
 - source commit:
14976b6c057549dcc00b5cd5020a97d6cfc7ffdb
 - receiver-source tree:
2db4aadfbbbe9bde6cf34cb20e98a638d0f9aad0f
- Restored receiver source inspected for this note: main commit 7fbdec9, especially `JunaCore/src/juna/common.jl` and `JunaCore/src/juna/frame_wide_ldpc.jl`.
- Existing regression test: `JunaCore/test/partial_fft_receiver.jl`.

The older legacy manifests do not record a source commit or tree, and the default files with four views do not print the view count in their CSV schema. Their directory pairs differ in the run wrapper by the explicit one-view setting, and OFDM results match exactly, but the missing provenance remains a reproducibility limitation.

8 Conclusion

We considered why four partial-FFT views can reduce pilot-only receiver performance at $N = 1024$ while improving JUNA-Lite. Summation with equal weights reconstructs the full FFT for every view count, but pilot-only Partial-FFT reception estimates unequal bandwise weights from limited pilots. JUNA-Lite then adds decoder posterior symbols as weighted anchors, unless a valid initial candidate stops the refinement.

The retained legacy results show a large JUNA-Lite increase with four views at code rate 0.125 and a large pilot-only decrease under impulsive added noise. They do not establish dependence on code rate, identify one physical cause, or validate the equality test with one view and one band spanning all carriers. The AWGN panel with four frames per path is also too small to order the receivers.

Under the stated 90-percent delay-window rule, CP128 is a conservative guard, while $N = 1024$ remains a tested choice for the simple OFDM receiver. The tap estimates alone do not establish that the replay channel stays fixed over the $N = 1024$ useful interval. The retained files therefore do not justify excluding JUNA-Lite. They show a descriptive four-view increase in the legacy rate-0.125 cases. Its AWGN improvement, code-rate dependence, cause, and processing trade-off remain undetermined until the controlled comparisons use matched frames and settings.

References

- [1] S. Yerramalli, M. Stojanovic, and U. Mitra, “Partial FFT Demodulation: A Detection Method for Highly Doppler Distorted OFDM Systems,” *IEEE Transactions on Signal Processing*, vol. 60, no. 11, pp. 5906–5918, November 2012.